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Voiced sounds — harmonic present!

⇒ but which amplitude observed in what is different for each alphabets

Energy of each word/alphabet is diff
Vocal cords - provide harmonics
• Who provides energy? — Shape of the vocal track

What harmonics amplify or attenuate, it decides

Vocal cord $\xrightarrow{harmonics}$ vocal track → that's how words differentiated

amplify or attenuate

• whisper — doesn't have harmonics
→ is harmonic used for understanding the speech?
⇒
If harmonic is used → then whisper is still used (only) differentiated by brain!!

Day 24 (27/03/26)
Q How do we segment the speech? (Phoneme)
JM
Smallest unit of sound which can change the word meaning

grapheme - Smallest unit of writing system

* for plotting spectogram → we req window size → if it's too small we won't notice the harmonics!!
Footnote: If we are able to see same ⇒ that's freq size < window size (formants)

Spectogram → narrowband (if window size large)

Spectogram → Wide band (If window size small)

↑ we may not spot harmonics

MFCC

Signal → extract features

- for long term features by statistical methods
- remove time and plot histogram

→ if non stationary data
  → take small window and process

→ Also note: report window wrt time and sampling rate !!

→ Spectogram is a matrix !

Spectogram → image → ML } wrong pipeline as image is full of pixels (0-255)

MFCC
↳ represents spectogram the way our brain processes

- hamming window - Shaped like gaussian

→ In lower freq take small bins and for higher freq take wider bins

as to distinguish higher freq is difficult

In frequency domain

and for amplitude, attenuate the higher freq amplitude !!

NOTE: between 2 consecutive frames, there is high correlation
↳ similar to 2 freq as well ← that we want to Reduce !!

→ to remove that correlation along columns → do DFT along (?)

⇓

Instead do DCT

↓ Some cosine trans

⇓ only!

This gives MFCC

→ Spectrogram ⟶ Small window ⟶ attenuation of amplitude } fig 1

large window ↓ over freq

↓

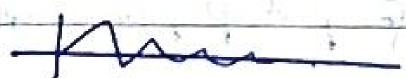
remove correlation } fig 2

(by DCT) ↓ MFCC !!

Day 25 (30/03/26)

- sine wave — amplitude, frequency, phase
- given any signal, can we represent it by k samples? (dimensionality reduction)

↓



↓ take FT

fill it with 0s } [c] k → just not first k we can take top k as well

↓ take IFT

> Can we have it like: we have FT such that in the transformed domain, the coefficients are already sorted?
>
> is there such a need? → for reducing dimension

This is discrete cosine transform (DCT)

This property is called energy ↓ coefficient compaction (Energy compaction) (~~sorting~~) !!

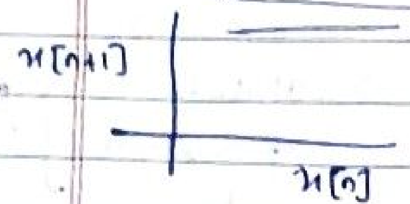
similar to PCA

Apply DCT to MFCC $\Rightarrow$ take first K

↑ that represents the feature vector

overall aim: reduce the dimension of the spectrogram

Linear Prediction : Time series data



↪ $X[n+1]$ is created by appending zero at the start ~~for~~ of $(x[n])$

↪ so just shifted!

- it represents the $x[n]$ and $x[n+1]$ is highly correlated i.e predictable

$$\left\{ x[n] = \underbrace{\sum_{k=1}^{p} a_k x[n-k]}_{\text{linear}} \right\}$$

← only this model doesn't generally holds up!!

$x[n]$ is summation for past p prev. samples

$$x[n] \; + = G \sum_{k=0}^{q} b_k \, e[n-k] \longrightarrow \text{(A)}$$

another time series

if we have this: then it represents real T.S

$$\cancel{\sum x[n] = \sum_{k=1}}$$

(Total # parameters = p+1)

$$\Rightarrow x[n] = \underbrace{\sum_{k=1}^{p} a_k x[n-k]}_{} + e[n]$$

$p << N$

$$x[n] \cong \sum_{k=1}^{p} a_k x[n-k]$$

error signal

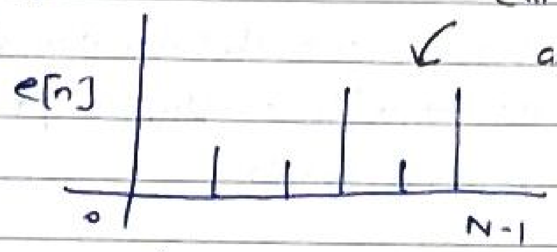
How to estimate a_k if $e[n]=0$?

M1 : minimise the error betn $x[n] - \sum_{k=1}^{p} a_k x[n-k]$

M2 :

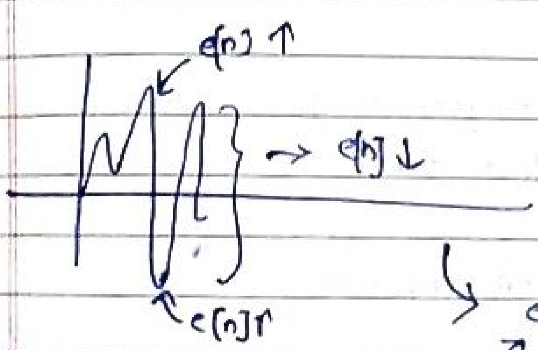
- for start elements → nothing is of past present
  ⇒ $e[n]$ should be high
- since $x[n] - \sum a_k x[n-k]$ will never go to 0,

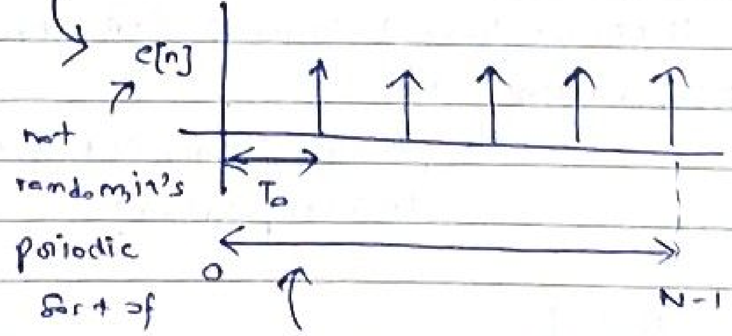
⇒



can be observed as train of impulses

⇒ non zero at some locations only





peak is observed at every T_0 sample

- if storing N samples ← sampling rate

⇒ $N \times 16 \text{ bits} \times F_s/\text{sec}$

bits per sec

each sample stored as 16 bits

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Since $e[n]$ not varying much

$$\Rightarrow \underbrace{p \times 16 \times F_s}_{\text{for } x[n]} + \underset{e[n]}{n} \times \boxed{\uparrow} \times F_s$$

less than 16 bits

Day 26 (02/04/26) # Modeling the eye (camera)

- light is also a time series
  (or sort of what our eyes see)
  ↳ but at a single time step, it is an image

- Detection threshold for green is low

- When we pass the image through a grid → it is sampling in spatial domain

~~Q~~ What is spatial?

- Grey scale — 0 means dark
  white / light → maximum presence of that colour.

- NOTE: we have 1 layer of RGB colours
  # Bayer filter ↓
  to get only Green, Red or Blue → we need to do interpolation!

we get this by passing only specific colour! → {Red, Green, Blue} matrices if added ⇒ gives grey scale image

Day 27 (06/04/26)

- tensor = matrix with another dimension
- Color info tells us the hight idea (it's actually a 3D plot)
- 8 bits used for resolution (0 to 255)

(dark) ← 0, 255 → White

(No colour)

- Grayscale Image advantage – image compression (from 3 channels → 1 channel)

disadvantage – particular channel might give more info than averaging all 3 channels

ex. Image recognition – skin tone (then grayscale not so good)

grayscale to binary image

binary → black and white image

→ check by threshold if > replace by 255 else 0

how to choose a threshold?

a) take avg

why avg? ≡

b) edge (intensity) preserved

pick threshold such that

c) depended on task as in what kind of values we want to preserve

d) draw the histogram → take mean or median value

windy.com

↳ weather data

- generally shown in Z-colourmap
- it's grayscale (but)

(how to convert the image !!)

e) decide threshold first → then 255, 0

→ go to random pixel → change the colour

random Dithering

f) floyd steinberg dithering → to preserve the edge, density of black pixels increase

shows sort of grayscale (but is actually binary image)

- Image augmentation – not just on rotation but intensity as well ↳ (noise additive augmentation)

How to encode colour information (without spending equal bits in all channels)

- (better than RGB)
- YP_bP_r model ← RGB transformed into

$Y \rightarrow$ intensity

$P_b, P_r \rightarrow$ colour info for blue and red

Grayscale – avg

Luminance – weighted avg ↑ (hardcoded values)

$$\begin{bmatrix} Y \\ P_b \\ P_r \end{bmatrix} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$ → (invertible)

⇒ from Y, P_b, P_r we can get R, G, B

⇒ note: humans more sensitive to intensity variation than colour

⇒ Y should be encoded better ← (by more no. of bits)

⇒ grain info intact there!

⇒ P_b, P_r can be encoded with lesser bits

but in RGB → for each matrix, we use 8 bits

- also alternate columns in P_b and P_r matrix can be thrown off
- as gradient is not perceptible!!

Illuminati

Day 28 (07/04/26)

- brain does collective processing (not pixel by pixel processing)
- 2 types of quantization
  - (i) intensity — no. of bits
  - (ii) spatial — how much should be the size
    - ↳ leads to spatial resolution
  - intensity resolution —
- blurring etc — are operations on an image
- how to capture binary images — pass a backlight and take pic

- pixel processing — going to a pixel and changing its value //
  - ↳ intensities can also transformed !!
  - ↳ binarization — gamma function

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Illuminati

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Day 29 (10/04/26)
- if edge present in small area ← detected better
  - sudden change in the magnitude of the signal
  - Specific to some direction !! ← it amplifies
- Robert — mask (45° more susceptible)
  - ↳ all values add up to 0 to avoid DC value
  - ↳ filter and mask same!

- Padding alternative ⇒ mirror symmetric
- black and white image → can be shown in colours (called heat map)

- Kernels generally considered symmetric so that during convolution, it can be implicitly flipped

- edge detection — as convolution
  - as fourier transform in freq domain

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- Time invariant process in image known as linear shift invariant system

- Box filter ≡ Rect function
  ↳ leads to saturation

- As we increase variance → the gaussian becomes flatter
  (for blurring the image)
  ⇓
  Smoothens the image

Day 30 (13/04/26)

- Convolution in image
  ↳ apply mask → (linear operation)
  ↓ (first flip mask → apply)
  replace the center value for each filter placed

- gaussian filter good as 2D filter can be split into sum of 2-1D filters

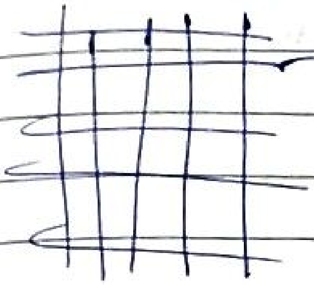
- Non-linear filters
  ↳ non linear operation - keeping a threshold

→ median filtering — noisy points (outliers) remove
  ↳ non convolution

→ template matching → similar to convolution
  ↳ here but subtraction done!!
  minimizing f cannot be done
  ↳ ~~cann~~ ⇒ maximise subtraction term
  ↳ which is convolution
  ↳ specifically cross correlation ~~convolution~~
  (filter not flipped)

- Convolution neural network — here filter is symmetric
  ⇓
  (i.e sort of similar to convolution
  else better is correlation) !!

- normalised C.C ⇒ helps in handling the energy of the signal



→ horizontal rows dominate the vertical axis in fourier domain

- is fourier transform of each channel (colour image) same ? (No)
- low pass filter ⇒ blurring
  high " " ⇒ sharpening

- how to apply high pass filter ?
  ↳ get the 2D fourier transform
  ↳ how to amplify the high frequency
  ↑
  far from zero

Images parts not imply

→ ~~high frequency~~ ~~signals~~ visible ~~filter~~ more frequencies to represent) required
early in transform

T1

T2:

Day 31 (17/04/26)

- spectral roll off!
- how brain ~~in~~ processes the signal
- how to remove noise from EEG?

⇓

→ Source separation pb
- → eyeblink detection
- → talker labelling – and what they were speaking

Day 32 (20/04/26)

- voltage gradient – results in signal generation
- how do humans learn?

↳ experiments → can it be modeled?

(neurons firing together get wired)

- how do babies learn?
- Why not MSE on a binary output? (model)

↳ Since derivative not possible (as min in the gradient dirn)

- is there a unique soln in perceptron algo?

↳ no depends on η

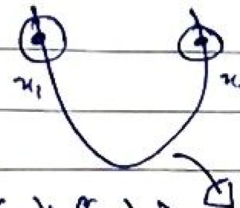
Converge only if data is linearly separable

what about DL?

Day 33 (21/04/26)

- Perceptron's converging not depended on learning rate
- how to test if a function is non convex?

↳ betn any two extreme points, the value is gonna be flipped



$f(x_1)+f(x_2) >$